Grade 11 – Term 1
C6–C7 Practice Activity with Solutions

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Evaluated criterion

C6. Estimates interval probabilities using relative frequencies from generated samples and compares them with theoretical probabilities.

C7. Calculates sample means and variances for different sample sizes and compares them with the theoretical mean and variance.

1 Introduction

Problems:

1.1. Generated samples, sample sizes, and histograms

1.2. Continuous distributions, theoretical and empirical probabilities

1.3. Sample mean and variance from generated data

Compare generated samples with a theoretical distribution using interval probabilities (C6), sample means, and sample variances (C7). Theoretical probabilities are areas under the probability density function (PDF); empirical probabilities are relative frequencies in the sample.

1.1 Real-world data, probability distributions, and histograms

A probability distribution models variation in a variable. A sample contains n collected or generated observations:

$$x_1, x_2, \ldots, x_n.$$

A histogram groups observations into intervals and shows their frequencies for comparison with the theoretical distribution.

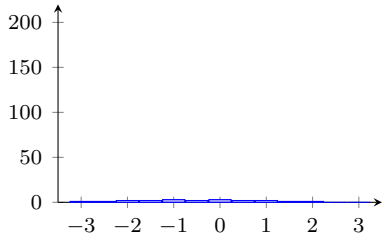


Figure 1: Count histogram from a sample of $n = 20$ observations.

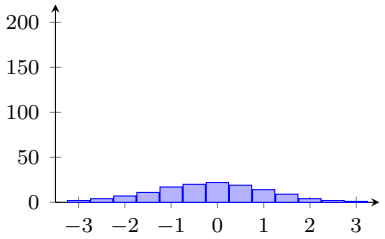


Figure 2: Count histogram from a sample of $n = 100$ observations.

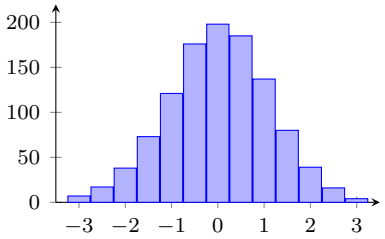


Figure 3: Count histogram from a sample of $n = 1000$ observations.

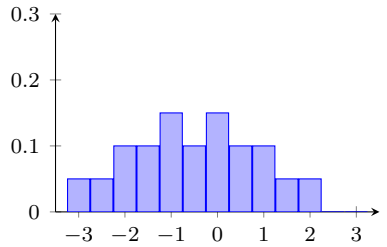


Figure 4: Histogram from a sample of $n = 20$ observations.

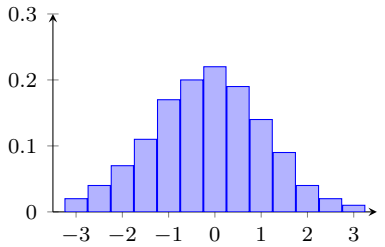


Figure 5: Histogram from a sample of $n = 100$ observations.

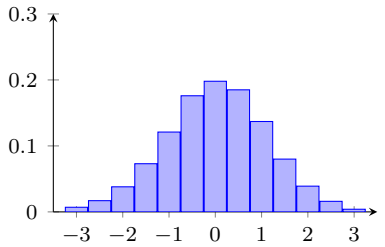


Figure 6: Histogram from a sample of $n = 1000$ observations.

Histograms vary between samples. Larger samples generally give a more stable approximation of the distribution's shape.

Compare the theoretical mean μ and variance σ^2 with the sample mean $\bar{x}$ and variance s^2:

Theoretical model	**Observed data**
Probability distribution	Observations $x_1, x_2, \ldots, x_n$
Theoretical mean μ	Sample mean $\bar{x}$
Theoretical variance σ^2	Sample variance s^2

Larger samples generally give more stable estimates of these parameters:

$$\bar{x} \approx \mu, \qquad s^2 \approx \sigma^2.$$

The relationship is

Real-world process $\longrightarrow$ Observed data $\longrightarrow$ Histogram and sample statistics $\longleftrightarrow$ Probability distribution



1.2 Continuous distributions, theoretical and empirical probabilities

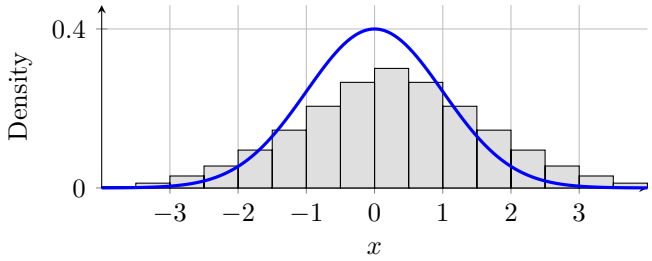


Figure 7: Theoretical normal probability density function and a histogram from a sample generated from the same distribution. The smooth curve represents the theoretical model, while the bars represent the observed sample.

For a continuous random variable, the theoretical probability of an interval $[a, b]$ is the area under its PDF:

$$P(a \le X \le b) = \int_a^b f(x)\,dx.$$

If k of n sample observations fall within the interval, the empirical probability is

$$\widehat{P}(a \le X \le b) = \frac{k}{n}, \qquad k = \#\{x_i : a \le x_i \le b\}.$$

Compare the two methods:

Theoretical probability	**Empirical probability**
Based on the probability distribution	Based on a generated sample
Area under the PDF	Relative frequency
$P(a \le X \le b) = \int_a^b f(x)\,dx$	$\widehat{P}(a \le X \le b) = \frac{k}{n}$

The graphs illustrate each method:

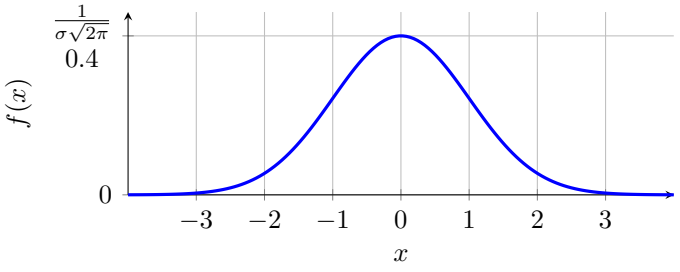


Figure 8: Theoretical probability density function of a normal distribution. The area under the curve over an interval represents the theoretical probability of that interval.

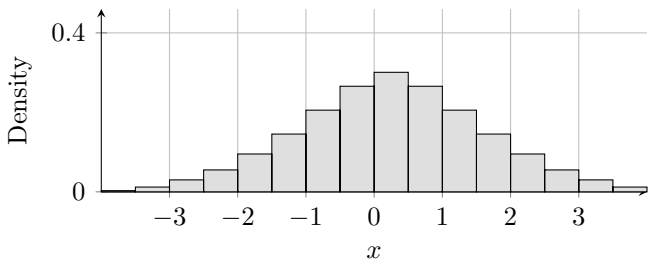


Figure 9: Histogram of a finite sample generated from the same distribution. The bars show the observed frequencies within the histogram bins.

In a histogram, observations within the selected interval determine k. For every distribution, calculate empirical probability using

$$\widehat{P} = \frac{\text{number of observations in the interval}}{\text{total number of observations}}.$$

Theoretical probability depends on the distribution's PDF:

$$\text{PDF area} \quad \longleftrightarrow \quad \text{histogram relative frequency}$$

These quantities generally become closer as sample size increases, though random sampling causes differences.

The difference between the empirical and theoretical probabilities is

$$\text{Difference} = \widehat{P}(a \le X \le b) - P(a \le X \le b).$$

The absolute difference is

$$\text{Absolute difference} = \left|\widehat{P}(a \le X \le b) - P(a \le X \le b)\right|.$$

When the theoretical probability is nonzero, the percentage error is

$$\text{Percentage error} = \frac{\left|\widehat{P}(a \le X \le b) - P(a \le X \le b)\right|}{P(a \le X \le b)} \times 100\%.$$



1.3 Sample mean and variance from generated data

Use the following notation for theoretical parameters and sample statistics:

Theoretical model	Generated sample
Probability distribution	Finite dataset
Mean μ	Sample mean $\bar{x}$
Variance σ^2	Sample variance s^2
Standard deviation σ	Sample standard deviation s

For ungrouped observations $x_1, x_2, \ldots, x_n$, the sample mean, variance, and standard deviation are

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

The mean measures the data's center; the variance measures spread around $\bar{x}$.

When observations are grouped into intervals, let f_i be the frequency of class i and m_i its midpoint:

$$m_i = \frac{a_i + b_i}{2}, \qquad n = \sum_i f_i.$$

The grouped-data mean and variance are

$$\bar{x} = \frac{\sum_i f_i m_i}{n}, \qquad s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1}.$$

Grouped calculations are approximate because all observations within an interval are represented by its midpoint.

Ungrouped data	Grouped data
$x_1, x_2, \ldots, x_n$	Intervals and frequencies
Exact observations	Class midpoints m_i
$\bar{x} = \frac{1}{n}\sum_i x_i$	$\bar{x} = \frac{\sum_i f_i m_i}{n}$
$s^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2$	$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1}$

For a finite random sample generated from a theoretical distribution,

$$\bar{x} \approx \mu, \qquad s^2 \approx \sigma^2.$$

Estimates vary between samples and generally become more stable as sample size increases.

2 C6 Practice: Empirical and Theoretical Probabilities

Problems:

2.1. Monthly household electricity use (raw data)

2.2. Waiting times (frequency data)

2.3. Package masses (grouped data)

The following problems focus only on estimating interval probabilities from samples. The larger samples make the relative-frequency estimates more stable, while the raw, frequency, and grouped formats provide different ways of organizing the observations.

2.1 Monthly household electricity use (raw data)

A utility company generates the following sample of monthly household electricity use (in hundreds of kWh):

3.8	4.1	4.5	5.2	5.7	6.0	6.3	6.8	7.1	7.5
4.0	4.4	4.9	5.4	5.9	6.1	6.5	6.9	7.3	8.0

For the theoretical model, $P(5 \le X \le 7) = 0.62$.

Questions/Tasks

C6 Estimate $P(5 \le X \le 7)$ from the sample. Then calculate the absolute error and the percentage error relative to the theoretical probability.

C6 Solution

There are $n = 20$ observations. The values in the interval from 5 to 7, inclusive, are

$$5.2,\ 5.7,\ 6.0,\ 6.3,\ 6.8,\ 5.4,\ 5.9,\ 6.1,\ 6.5,\ 6.9,$$

so $k = 10$. Therefore,

$$\widehat{P}(5 \le X \le 7) = \frac{k}{n} = \frac{10}{20} = 0.50.$$

The absolute error is

$$E = |0.50 - 0.62| = 0.12,$$

and the percentage error is

$$\frac{|0.50 - 0.62|}{0.62} \times 100\% = \frac{0.12}{0.62} \times 100\% \approx 19.35\%.$$

Thus the empirical probability is 0.50, which is 0.12 below the theoretical probability.



2.2 Waiting times (frequency data)

A clinic records the waiting times of 120 patients in the following frequency table:

Waiting time (min)	5	10	15	20	25	30
Frequency	8	20	34	30	18	10

For the theoretical model, $P(10 \le X \le 20) = 0.72$.

Questions/Tasks



C6 Estimate the probability that a patient waits from 10 to 20 minutes, inclusive. Calculate its absolute error and compare the empirical estimate with the theoretical probability.

C6 Solution

The total frequency is

$$n = 8 + 20 + 34 + 30 + 18 + 10 = 120.$$

The waiting times 10, 15, and 20 minutes are in the interval, so

$$k = 20 + 34 + 30 = 84.$$

Hence

$$\widehat{P}(10 \le X \le 20) = \frac{84}{120} = 0.70.$$

The absolute error is

$$E = |0.70 - 0.72| = 0.02.$$

Thus the empirical probability is 0.70, which is 0.02 below the theoretical probability of 0.72.



2.3 Package masses (grouped data)

A distribution center groups the masses of 200 packages as follows:

Package mass (kg)	0–2	2–4	4–6	6–8	8–10
Frequency	18	46	72	44	20

The classes are left-inclusive and right-exclusive. For the theoretical model, $P(2 \le X < 8) = 0.83$.

Questions/Tasks



C6 Estimate $P(2 \le X < 8)$ from the grouped sample. Calculate the absolute error and the percentage error relative to the theoretical probability.

C6 Solution

The total frequency is

$$n = 18 + 46 + 72 + 44 + 20 = 200.$$

The interval $2 \le X < 8$ contains the three classes 2–4, 4–6, and 6–8. Their combined frequency is

$$k = 46 + 72 + 44 = 162.$$

Therefore,

$$\widehat{P}(2 \le X < 8) = \frac{162}{200} = 0.81.$$

The absolute error is

$$E = |0.81 - 0.83| = 0.02,$$

and the percentage error is

$$\frac{|0.81 - 0.83|}{0.83} \times 100\% = \frac{0.02}{0.83} \times 100\% \approx 2.41\%.$$

Thus the empirical estimate is 0.81, which is close to, but slightly below, the theoretical probability of 0.83.



3 C7 Practice: Sample Mean, Variance, and Standard Deviation

Problems:

3.1. Machine calibration errors (raw data)

3.2. Books borrowed (frequency data)

3.3. Commute times (grouped data)

The samples in this section are deliberately small so that the sample mean, sample variance, and sample standard deviation can be calculated clearly by hand. Recall that the sample-variance denominator is $n - 1$.

3.1 Machine calibration errors (raw data)

The calibration errors (in tenths of a millimeter) from five trials are

$$4, \quad 5, \quad 6, \quad 7, \quad 8.$$

The theoretical model has $\mu = 6$ and $\sigma^2 = 2.25$.

Questions/Tasks



C7 Calculate the sample mean, sample variance, and sample standard deviation. Compare the sample mean and variance with the theoretical values.

C7 Solution

There are $n = 5$ observations. First, calculate the sample mean:

$$\bar{x} = \frac{4 + 5 + 6 + 7 + 8}{5} = \frac{30}{5} = 6.$$

Next, calculate the squared deviations from $\bar{x} = 6$:

x_i	4	5	6	7	8
$x_i - \bar{x}$	-2	-1	0	1	2
$(x_i - \bar{x})^2$	4	1	0	1	4

Their sum is $4+1+0+1+4=10$. Therefore, the sample variance is

$$s^2 = \frac{\sum_{i=1}^{5}(x_i-\bar{x})^2}{n-1} = \frac{10}{5-1} = 2.50.$$

The sample standard deviation is

$$s = \sqrt{s^2} = \sqrt{2.50} \approx 1.58.$$

Thus,

$$\bar{x} = 6 = \mu, \qquad s^2 = 2.50 \approx \sigma^2 = 2.25, \qquad s \approx 1.58.$$



3.2 Books borrowed (frequency data)

The numbers of books borrowed by four students are summarized below:

x_i	2	4	6
f_i	1	2	1

The theoretical model has $\mu = 4$ and $\sigma^2 = 3$.

Questions/Tasks



C7 Calculate the sample mean, sample variance, and sample standard deviation from the frequency table. Compare the sample mean and variance with the theoretical values.

C7 Solution

The total frequency, and hence the sample size, is

$$n = 1+2+1 = 4.$$

The sample mean is

$$\bar{x} = \frac{\sum_i f_i x_i}{n} = \frac{1(2)+2(4)+1(6)}{4} = \frac{16}{4} = 4.$$

Using $\bar{x} = 4$, calculate each weighted squared deviation:

x_i	2	4	6
f_i	1	2	1
$(x_i-\bar{x})^2$	4	0	4
$f_i(x_i-\bar{x})^2$	4	0	4

The sum of the weighted squared deviations is $4 + 0 + 4 = 8$. Thus,

$$s^2 = \frac{\sum_i f_i (x_i - \bar{x})^2}{n - 1} = \frac{8}{4 - 1} = \frac{8}{3} \approx 2.67.$$

The sample standard deviation is

$$s = \sqrt{s^2} = \sqrt{\frac{8}{3}} \approx 1.63.$$

Therefore,

$$\bar{x} = 4 = \mu, \qquad s^2 \approx 2.67 \approx \sigma^2 = 3, \qquad s \approx 1.63.$$



3.3 Commute times (grouped data)

The commute times of five employees are grouped into three classes:

Commute time (min)	0–10	10–20	20–30
Frequency	1	3	1

The theoretical model has $\mu = 15$ minutes and $\sigma^2 = 48$ minutes^2.

Questions/Tasks



C7 Use class midpoints to estimate the sample mean, sample variance, and sample standard deviation. Compare the estimated mean and variance with the theoretical values.

C7 Solution

First, calculate each class midpoint:

$$m_1 = \frac{0 + 10}{2} = 5, \qquad m_2 = \frac{10 + 20}{2} = 15, \qquad m_3 = \frac{20 + 30}{2} = 25.$$

The total frequency is

$$n = 1 + 3 + 1 = 5.$$

Using the midpoints as representative values, the estimated sample mean is

$$\bar{x} = \frac{\sum_i f_i m_i}{n} = \frac{1(5) + 3(15) + 1(25)}{5} = \frac{75}{5} = 15 \text{ min}.$$

Next, calculate the weighted squared deviations:

m_i	5	15	25
f_i	1	3	1
$(m_i - \bar{x})^2$	100	0	100
$f_i(m_i - \bar{x})^2$	100	0	100

Their sum is $100 + 0 + 100 = 200$. The estimated sample variance is

$$s^2 = \frac{\sum_i f_i(m_i - \bar{x})^2}{n - 1} = \frac{200}{5 - 1} = 50 \text{ min}^2.$$

The estimated sample standard deviation is

$$s = \sqrt{s^2} = \sqrt{50} \approx 7.07 \text{ min}.$$

Therefore,

$$\bar{x} = 15 \text{ min} = \mu, \qquad s^2 = 50 \text{ min}^2 \approx \sigma^2 = 48 \text{ min}^2, \qquad s \approx 7.07 \text{ min}.$$

These results are estimates because all observations in a class are represented by the class midpoint.

4 Samples, Empirical Probabilities, and Sample Statistics

Problems:

4.1. Investment returns (raw data)

4.2. Daily energy consumption (frequency data)

4.3. Delivery times (grouped data)

4.4. Library visit durations (histogram data)

The following examples use small samples so that each step of the calculations can be shown clearly. In real-world applications, sample sizes are typically much larger to obtain more reliable estimates.

4.1 Investment returns (raw data)

A financial analyst records these returns (in percent) over six days:

$$2, \quad 2, \quad 3, \quad 3, \quad 4, \quad 4.$$

The theoretical model has $\mu = 3$, $\sigma^2 = 1$, and $P(2 \le X \le 4) = 0.80$.

Questions/Tasks

C6 Calculate the empirical probability of a return between 2% and 4%, inclusive, and its absolute error.

C7 Calculate the sample mean and sample variance, and compare them with the theoretical values.

C6 Solution

All six observations are in the interval, so $n = 6$ and $k = 6$. Hence

$$\widehat{P}(2 \le X \le 4) = \frac{k}{n} = \frac{6}{6} = 1.$$

The absolute error is

$$E = |\widehat{P} - P| = |1 - 0.80| = 0.20.$$

Thus the empirical probability is 1, which is 0.20 above the theoretical probability.



C7 Solution

The sample mean is

$$\bar{x} = \frac{2+2+3+3+4+4}{6} = \frac{18}{6} = 3.$$

Using $\bar{x} = 3$, the sample variance is

$$\begin{aligned} s^2 &= \frac{(2-3)^2 + (2-3)^2 + (3-3)^2 + (3-3)^2 + (4-3)^2 + (4-3)^2}{6-1} \\ &= \frac{1+1+0+0+1+1}{5} = \frac{4}{5} = 0.80. \end{aligned}$$

Therefore,

$$\bar{x} = 3 = \mu, \qquad s^2 = 0.80 \approx \sigma^2 = 1.$$



4.2 Daily energy consumption (frequency data)

A facility records daily consumption (in MWh) as follows:

x_i	8	10	12
f_i	1	2	1

The theoretical model has $\mu = 10$, $\sigma^2 = 3$, and $P(10 \le X \le 12) = 0.60$.

Questions/Tasks



C6 Calculate the empirical probability of consumption between 10 and 12 MWh, inclusive, and its absolute error.

C7 Calculate the sample mean and sample variance, and compare them with the theoretical values.

C6 Solution

The total frequency and the frequency in the interval are

$$n = 1 + 2 + 1 = 4, \qquad k = 2 + 1 = 3.$$

Therefore,

$$\widehat{P}(10 \le X \le 12) = \frac{3}{4} = 0.75,$$

and the absolute error is

$$E = |0.75 - 0.60| = 0.15.$$

Thus the empirical probability is 0.75, which is 0.15 above the theoretical probability.



C7 Solution

The sample mean is

$$\bar{x} = \frac{1(8) + 2(10) + 1(12)}{4} = \frac{40}{4} = 10.$$

The sample variance is

$$\begin{aligned} s^2 &= \frac{1(8-10)^2 + 2(10-10)^2 + 1(12-10)^2}{4-1} \\ &= \frac{4+0+4}{3} = \frac{8}{3} \approx 2.67. \end{aligned}$$

Therefore,

$$\bar{x} = 10 = \mu, \qquad s^2 \approx 2.67 \approx \sigma^2 = 3.$$



4.3 Delivery times (grouped data)

A company groups delivery times into three classes:

Delivery time (min)	10–20	20–30	30–40
Frequency	1	2	1

The theoretical model has $\mu = 25$ minutes, $\sigma^2 = 75$ minutes^2, and $P(20 \le X < 30) = 0.45$.

Questions/Tasks

C6 Estimate the empirical probability of a delivery time from 20 up to (but not including) 30 minutes, and calculate its absolute error.

C7 Estimate the sample mean and sample variance using class midpoints, and compare them with the theoretical values.

C6 Solution

The total frequency is $n = 1 + 2 + 1 = 4$. The selected class has frequency $k = 2$, so

$$\widehat{P}(20 \le X < 30) = \frac{2}{4} = 0.50.$$

The absolute error is

$$E = |0.50 - 0.45| = 0.05.$$

Thus the empirical probability is 0.50, which is 0.05 above the theoretical probability.

C7 Solution

The three class midpoints are

$$15, \qquad 25, \qquad 35.$$

Using these midpoints, the estimated sample mean is

$$\bar{x} = \frac{1(15) + 2(25) + 1(35)}{4} = \frac{100}{4} = 25 \text{ min}.$$

The estimated sample variance is

$$\begin{aligned} s^2 &= \frac{1(15-25)^2 + 2(25-25)^2 + 1(35-25)^2}{4-1} \\ &= \frac{100 + 0 + 100}{3} = \frac{200}{3} \approx 66.67 \text{ min}^2. \end{aligned}$$

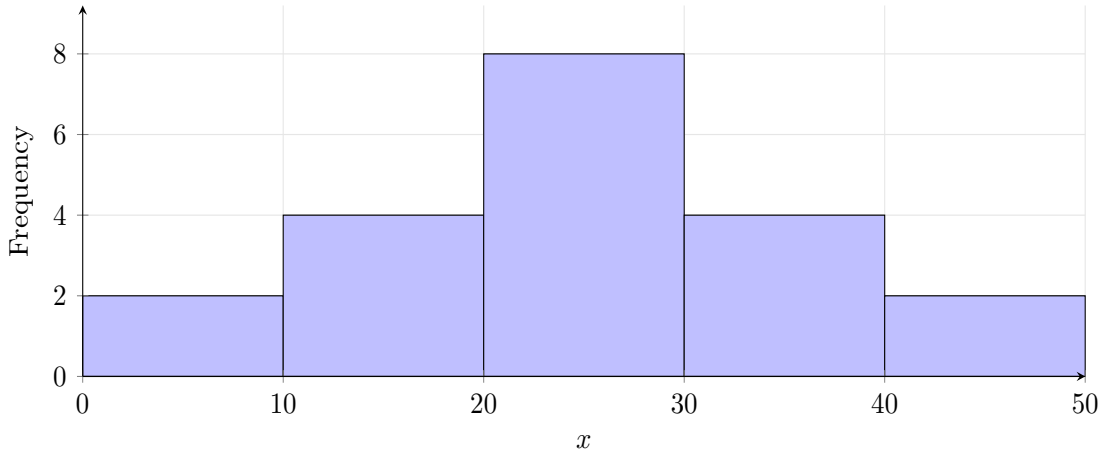
Therefore,

$$\bar{x} = 25 \text{ min} = \mu, \qquad s^2 \approx 66.67 \text{ min}^2 \approx \sigma^2 = 75 \text{ min}^2.$$

These grouped-data results are estimates because every observation in a class is represented by its midpoint.

4.4 Library visit durations (histogram data)

A school librarian records the duration x, in minutes, of each visit in a sample of student visits. The histogram below is the only record provided of the sample. Each bar shows the number of visits in its class; classes are left-inclusive and right-exclusive.



Visit duration x (minutes); vertical axis: frequency (number of visits).

Questions/Tasks

C6 Read each class interval and its frequency from the histogram, and determine the total sample size. Calculate the empirical probabilities $\widehat{P}(20 \le X < 30)$ and $\widehat{P}(10 \le X < 40)$. Find the cumulative frequency and cumulative empirical probability below each upper class boundary.

C7 Use class midpoints to estimate the sample mean, sample variance, and sample standard deviation. Explain why these statistics are estimates.

C6 Solution

The horizontal axis gives the class boundaries, and the vertical axis has grid spacing 2. Reading the bar heights gives

Visit duration (min)	$[0, 10)$	$[10, 20)$	$[20, 30)$	$[30, 40)$	$[40, 50)$
f_i	2	4	8	4	2

The total frequency is the sample size:

$$n = \sum_i f_i = 2 + 4 + 8 + 4 + 2 = 20.$$

The class $[20, 30)$ has frequency 8, so

$$\widehat{P}(20 \le X < 30) = \frac{8}{20} = 0.40.$$

The interval $[10, 40)$ contains three complete classes. Therefore,

$$k = 4 + 8 + 4 = 16, \qquad \widehat{P}(10 \le X < 40) = \frac{16}{20} = 0.80.$$

For each upper boundary b, add the frequencies of all classes below b. Divide that cumulative frequency by $n = 20$ to obtain $\widehat{P}(X < b)$:

b (min)	Cumulative frequency	$\widehat{P}(X < b)$
10	2	$2/20 = 0.10$
20	$2 + 4 = 6$	$6/20 = 0.30$
30	$2 + 4 + 8 = 14$	$14/20 = 0.70$
40	$2 + 4 + 8 + 4 = 18$	$18/20 = 0.90$
50	$2 + 4 + 8 + 4 + 2 = 20$	$20/20 = 1$

The last cumulative frequency equals the sample size, and the last cumulative probability is 1, as expected.



C7 Solution

First, calculate each midpoint using $m_i = (a_i + b_i)/2$. For example, the midpoint of $[0, 10)$ is $(0 + 10)/2 = 5$ minutes. The midpoints and weighted values are

m_i	5	15	25	35	45
f_i	2	4	8	4	2
$f_i m_i$	10	60	200	140	90

The estimated sample mean is

$$\bar{x} = \frac{\sum_i f_i m_i}{n} = \frac{10 + 60 + 200 + 140 + 90}{20} = \frac{500}{20} = 25 \text{ min}.$$

Next, use $\bar{x} = 25$ to calculate the weighted squared deviations:

m_i	5	15	25	35	45
f_i	2	4	8	4	2
$(m_i - \bar{x})^2$	400	100	0	100	400
$f_i(m_i - \bar{x})^2$	800	400	0	400	800

Their sum is $800 + 400 + 0 + 400 + 800 = 2400$. Using the sample-variance denominator $n - 1$, the estimated variance and standard deviation are

$$s^2 = \frac{\sum_i f_i(m_i - \bar{x})^2}{n - 1} = \frac{2400}{20 - 1} = \frac{2400}{19} \approx 126.32 \text{ min}^2,$$

$$s = \sqrt{s^2} = \sqrt{\frac{2400}{19}} \approx 11.24 \text{ min}.$$

Thus the midpoint estimates are

$$\bar{x} = 25 \text{ min}, \qquad s^2 \approx 126.32 \text{ min}^2, \qquad s \approx 11.24 \text{ min}.$$

These statistics are estimates because the histogram does not give the individual durations: every observation in a class is represented by its midpoint. The class frequencies and the empirical probabilities for unions of complete classes, however, are read exactly from the histogram.

